P-192: calibrated first-coefficient decoding and horizontal transport

One reusable calibration + seven first spectral coefficients

Bcal = B(P,P); y_i = Bcal*y(P)/B(P+iG,P+iG+U). No C or c_Q input.

Cubic candidates -> unique signed point

At most three x roots; two distinct candidates cannot agree at seven shifts (pole degree six).

Horizontal transport retains its derivative correction

C'/B' = F(C/B); B' = H B / F'(C/B). Require 2f(P) nonrational.

Actual P192 transfer -> supplied maps and explicit inverse

47 maps and closed duals; V = [degree^(-1) mod r] dual(f)(V_destination).

Construction, 2026-11-08, T-1942. A horizontal isogeny can now transport the marked norm coefficients and decoded point with an explicit derivative correction. The multiplier of all divided differences has a short Frobenius-quotient expression, while the target-specific transverse mark is retained. The accompanying bidirectional point-action reduction uses the dual isogeny and one known scalar inverse, with no enumeration of geometric preimages. Written derivations and native checks remain producer evidence. [pcc:unverified]

Calibrated decoder. A new written reduction needs only B, the coefficient of a uniform spectral shift of the graph factors. First query Bcal=B_Q(P,P). For i=0,...,6, query B_i at shift P+iG with the same U. Then y_i=Bcal*y(P)/B_i equals y(P+jG+iG), where [j](G-Q)=U. One cubic has at most three point candidates; checking these seven y values selects exactly one, because the difference of two translated y functions has pole degree six. B-only-proof.md gives the general proof, public exceptions and costs. [pcc:unverified]

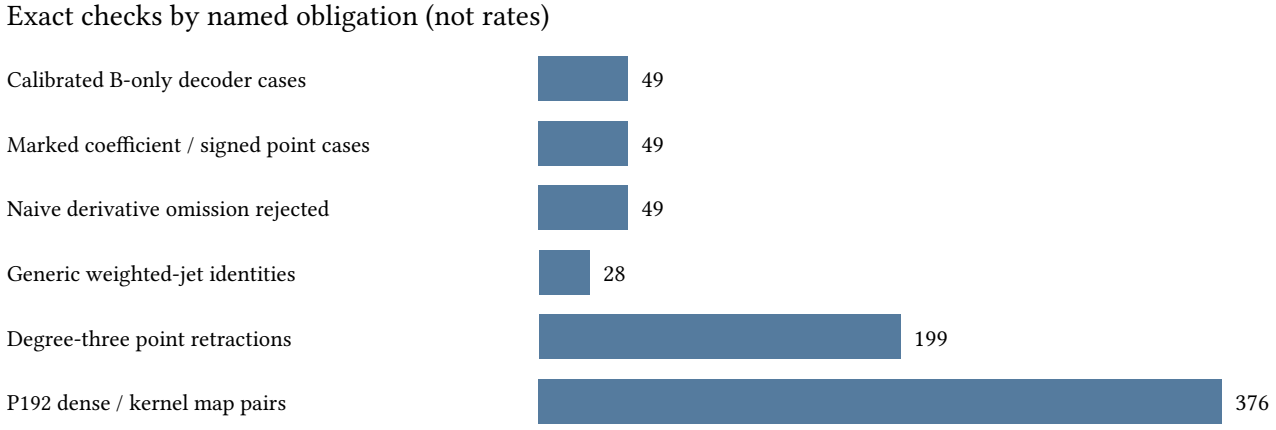
Decoder qualification. Both amendments were published before the B-only mode ran. All 49 declared cases recover through the calibrated formula without computing C. Streaming (A,B) and independent prefix/suffix outputs agree. Each of 49 changed y transcripts is passed through the actual root decoder and rejected. Seven reusable calibration queries and 343 shifted queries contract 69,650 factors. Native cubic solving explicitly enumerates 2,181,529 field values across the original cases and the same number on negative controls; this is not assigned a P192 cost. [pcc:unverified]

Exact interface. For an F_p-isogeny with x-map F, define h(X,Y)=(F(X)-F(Y))/(X-Y), using F'(X) on the diagonal. The product H of h over the coupled graph equals (c'/c)*h(x((1-pi)P),x((1-pi)R)). On R=P+U with 2f(P) nonrational, the marked coefficients obey C'/B'=F(C/B) and B'=H*B/F'(C/B). The source and destination have the same unique vanishing factor. The proof includes all guards, nonuniform weights, and homogeneous ratio transport. The normalization constants c,c' are still charged setup. [pcc:unverified]

Implemented coefficient controls. The exact degree-three control is E/F211:y^2=x^3+7 -> E'/F211:y^2=x^3+22, with F(x)=x+28/x^2 and y'=y*(1-56/x^3). Both rational groups have declared order199; full source point enumeration by the inherited generator and every destination image are checked. The kernel is O and (0,+/-sqrt(7)), defined over the quadratic extension with Frobenius-stable subgroup. This is a ramified CM control, not a split edge on P192. A frozen lexicographic shift scan chooses P=(1,25W), W^2=210, with f(P)=(29,102W), before coefficient comparisons. [pcc:unverified]

Positive and negative paths. All 49 declared target/input pairs preserve the source coefficient ratio, destination coefficient formula, signed lift and dual retraction. Each source/destination coefficient contraction agrees with a separate prefix/suffix cofactor implementation. All 49 reject the naive B'=H*B formula omitting F'(C/B). Twelve nonzero sign controls keep Q and -Q distinct. Twenty-eight generic input pairs check the entire weighted square-zero identity and compact multiplier, and all 199 rational points pass the degree-three map/retraction identity. These are deterministic declared cases, without sampling, timing or reliability inference. [pcc:unverified]

P-192 map transport. The full-width native checker consumes the actual Round27 47-edge prefix and Round28 closed duals. It transports G, the existing unknown public Q, G-Q and 2G, checks curve membership at every edge, compares dense rational maps with independent compact kernel evaluation, and retracts using the product degree inverse modulo r. All 376 map pairs agree, and the transported difference/doubling relations agree. No target scalar is supplied. These are point-map checks, not a full P192 norm evaluation. [pcc:unverified]



Sources: B-only.json, transport.json, P192-transport.json. Unit: exact checks; deterministic declared cases.

Cost inventory. The complete two-coefficient transfer control makes 154 ordinary full norm contractions and 77 weighted contractions, totaling 45,969 contracted factors. It also multiplies 15,323 h factors and 2,772 constant-normalization terms; public point operations, chart tests, h evaluations, root/lift search and reference/fresh replays are additional work. Two G graphs contain 398 points and 14 target graphs contain 2,786 points. The P192 route uses 94 maps per public point per representation, reads 9,752 dense coefficients per point, and pays the known 192-bit degree-inverse scalar multiplication separately. These counts are not online time intervals or a claim of fast initial jet evaluation. [pcc:unverified]

Actual CM relation. The replay recomputes the product of the 53 prime degrees as the retained 292-bit norm and their sum as1,151,567. Its 47-map prefix has 205-bit degree; six global maps remain unconstructed here. A literal full fiber expansion has the product degree as its branch count, whereas a supplied chain evaluates points and represented jets through the sum of local map costs. The dual reduction avoids that geometric branch expansion for point-action transfer. It does not replace a missing global map by its rational-subgroup scalar correction. [pcc:unverified]

Compression consequence. Given a source coefficient pair, its projective image and absolute scaling can follow the chain without expanding the composite map. Given a destination marked-action evaluator, dual plus degree inverse pulls the answer back. This makes endpoint selection and CM reduction usable without discarding the marked information. The initial public-input coefficient evaluator remains a separate circuit obligation: the compact multiplier and closed-chain linearity do not seed C/B. The calibrated decoder sharpens that task to a normalized spectral first response B(P,P+U)/B(P,P). Next, investigate a target-specific circuit for that response or a subset representation that can initialize it, with all construction costs included. [pcc:unverified]

Validation and custody. Native source and exact scope were committed and published before affected checks. Five final fresh-copy outputs and mutation controls are recorded in validation.json; the required legacy library check is recorded with its actual exit. Source, destination, references, negative cases and all scan rejects remain in the bundle. The editable flow, quantitative graph and inspected PDF describe this round only. No cloud job, managed generator or old source freeze lane was started or modified. [pcc:unverified]

Sources checked. The live Round60/61 norm and transporter proofs and Round25/27/28 relation/map files supply the local construction and inherited assumptions. BFLS, https://arxiv.org/pdf/2003.10118 , section4.15 permits isogeny-composed coordinates. Banegas et al, https://arxiv.org/pdf/2306.16072 , sections2-3 and5.2 separate isogeny/kernel evaluation and Frobenius savings. Robert HDR, https://www.normalesup.org/~robert/pro/publications/academic/hdr_details.pdf , section4.7.3 separates compact description from reconstruction. Initial arxiv HTML fetches returned404; the primary PDFs were read. KB tools/wrapper remain unavailable. No novelty or absence claim is inferred from the search. [pcc:unverified]